

M Y W O R K

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Systems design and scientific literature — production architecture, published research, and anonymized client engagements. Everything below is live, peer-reviewed, or delivered.

Neural-State Ventilation Engine

Client: Project Phrenic Edge · Critical Care · Edge AI & Closed-Loop Control

Closed-loop AI architecture for brainstem-driven ventilation. Classifies eight pathological breathing patterns in real time, applies non-invasive phrenic nerve stimulation, and maintains a deterministic <50ms safety fallback. Full SaMD regulatory strategy across five jurisdictions.

8

ML PATTERN CLASSES

<50ms

SAFETY FALLBACK

5

JURISDICTIONS

[Full Case Study →](#)

Client Engagements — Delivered Private Consulting

Anonymized case studies published with written client consent. No proprietary or trade secrets are disclosed:

Early Signal Monitor

Client: DAX-40 Pilot

Predictive health monitoring architecture for a large industrial workforce. Identifies clinical stress signals months before the first long-term absence case.

- Focus: **Scalability**

[Case Study](#)

Patient Platform Engine

Client: Private Practice

Six-language patient journey architecture for a private practice expanding from pediatrics to adult medicine. Cross-cultural clinical care as an engineering problem, not a translation one.

- Focus: **Expansion**

[Case Study](#)

Engagement Timing Model

Client: Voice Platform

Algorithmic timing engine optimizing patient communication by matching delivery to neural receptivity windows. Adherent communication, not just more communication.

- Focus: **Patient Journey**

[Case Study](#)

Neural-State Ventilation Engine

Client: Project Phrenic Edge

Closed-loop AI architecture for brainstem-driven ventilation. Eight pathological breathing patterns classified in real time with deterministic <50ms safety fallback.

- Focus: **Critical Care**

[Case Study](#)

Research Systems — Open Source, Ongoing

Production-ready architecture deployed and accessible online:

Clinical Safety Gate

Repository: CGNR Matrix

Deterministic logic engine validating AI outputs against hard-coded clinical protocols prior to patient delivery.

- Focus: **Patient Safety**

[Live](#) · [GitHub](#)

Decision Engine

Repository: MOAC QMM

Algorithmic decision engine processing complex pharmacological interactions and high-density physiological edge cases.

- Focus: **Clinical Validity**

[Live](#) · [GitHub](#)

Compliance Shield

Repository: RAGNAR ⚡ OMNI

Automated regulatory engine fencing clinical outputs within the correct legal perimeter per jurisdiction. Protects the company, not just the patient.

- Focus: **Governance**

[Live](#) · [GitHub](#)

Lab Data Guardian

Repository: ChronosLab

Anti-drift governance for medical lab parameters — normalizes reference ranges across labs and flags values outside plausible clinical bounds.

- Focus: **Data Integrity**

[Live](#) · [GitHub](#)

Peer-Reviewed Publications

Sustained participation and quality of life after three decades of phrenic nerve stimulation and home mechanical ventilation in high cervical spinal cord injury: a clinical case report

Laura Peiler, Nabil Ktari, Maria-Cristina Hallwachs, Sven Hirschfeld, Martin Groß

Springer Nature · Spinal Cord · Case Report · Accepted 2026 · Impact Factor 2.5

Objective: Healthcare professionals generally underestimate quality of life (QOL) after high cervical spinal cord injury (SCI) with ventilator dependence compared with patient self-reports. This case provides long-term observational data from a community-dwelling individual with high cervical SCI.

Methods: Clinical case report and semi-structured interview.

Results: A woman sustained C1–C3, AIS A tetraplegia from a diving accident at age 18. Bilateral phrenic nerve stimulator implantation at one year post-injury permitted daytime ventilator-free breathing and physiological speech. Now aged 50, she has held leadership positions in national organizations, chairs a charitable foundation, and travels internationally. She reported high life satisfaction, describing her work as both exhausting and meaningful.

Conclusion: Complete high cervical tetraplegia with ventilator dependence was compatible with three decades of community living, professional leadership, and reported well-being. Factors associated with this outcome included community-based living, family resources, the ability to speak, and sustained engagement in meaningful activity.

[Full Publication Summary →](#)

Profil épidémioclinique, histologique et thérapeutique du syndrome de Sweet : à propos de 25 cas

S. Harbaoui, N. Litaïem, M. Jones, C. Chouk, N. Ktari, S. Rammeh, N. Ezzine, F. Zeglaoui

Elsevier · Annales de Dermatologie et de Vénéréologie · Published May 2018

DOI: 10.1016/j.annder.2018.03.049

Introduction: Le syndrome de Sweet (SS), décrit par le Dr Robert Douglas Sweet en 1964, appartient au groupe des dermatoses neutrophiliques.

But: Étudier les particularités épidémiocliniques, histologiques et thérapeutiques observées au cours du SS à travers une série hospitalière.

Méthodes: Étude rétrospective, monocentrique, descriptive, colligeant 25 malades atteints du SS et hospitalisés entre janvier 1999 et décembre 2015.

Résultats: Vingt-cinq patients inclus. Âge moyen 50,5 ans. Sex-ratio F/H = 5,25. Stade prodromique chez 19 patients. Plaques érythémato-papuleuses infiltrées chez tous les malades. À l'histologie, infiltrat inflammatoire riche en PNN dans tous les cas. AINS en première ligne (72%), corticoïdes systémiques (12%), dermocorticoïdes (8%), colchicine (4%).

Conclusion: Le SS fait partie du spectre des maladies neutrophiliques. L'association à une étiologie néoplasique justifie la réalisation d'un bilan étiologique clinique et paraclinique initial.

DOI: 10.1016/j.annder.2018.03.049 →